Vincent S. Pronel

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Education

Bachelor of Science in Computer Science **Expected 2027**
Kean University Union, NJ

- **Relevant Coursework**: Applied Statistics, Matrix & Linear Algebra, Analysis of Algorithms*, Database Management Systems, Software Engineering*, Computer Architecture* **in progress, Fall 2026*
- **Leadership & Honors**: Dean's List, Spring 2024 and Fall 2025 | Presenter, 2026 CRA UR2PhD Research Showcase, New Orleans | GMiS Research Posters Competition | Web Developer, WiCyS (Women in Cyber Security) Kean Chapter E-Board

Technical Skills

- **AI & Evaluation**: OpenAI and Google Gemini APIs, supervised fine-tuning pipelines, JSONL dataset preparation, LLM-as-a-judge evaluation, prompt design, human-subject study design
- **Data & Analysis**: SQL, MySQL, PostgreSQL, Supabase, correlation analysis, data aggregation and normalization, Google Charts
- **Languages**: Python, Java, TypeScript, JavaScript, SQL, PHP, HTML/CSS
- **Engineering & Tools**: Next.js, React, Node.js, REST APIs, JDI, JavaParser, Cloudflare (Workers, D1, R2), Vercel, Git/GitHub, Linux

Research & Engineering

AI-Assisted Java Execution Visualizer | *CRA UR2PhD-REU Fellowship* June 2026 – Present
Undergraduate Research Fellow, Advisor: Dr. Yan Ma | Competitively selected, $8,000 funded, 400 hours *Kean University*

- Independently built a Next.js/TypeScript Java memory visualizer with 4 guided simulations and 26 execution states, representing stack frames, heap objects, arrays, aliasing, method calls, and reference changes line by line.
- Normalized raw JDI execution traces into a typed visualization model by reconciling pre-execution events, suppressing constructor internals, renumbering JVM object IDs, deduplicating states, and deriving reference graphs and per-step diffs.
- Implemented an IRB-approved 2-condition experiment targeting 40 completed sessions, capturing pre/post responses, elapsed time, lesson behavior, and an immutable 11-event audit trail for paired analysis.

Jobs vs. Wages | *NSF IUSE Scholars Academy* May 2026 – Present
NSF IUSE Research Scholar, Advisor: Dr. Pauline Wang | Competitively selected, $1,400 funded, 10 weeks *Kean University*

- Built a MySQL data mart from 33,000 records in the US ZIP Codes Database and wrote population-weighted aggregation logic for all 50 states, Washington, D.C., and Puerto Rico; presented the findings at the 2026 GMiS conference in Albuquerque.
- Disproved the study's own premise: unemployment explains only 14% of variation in household income ($r = -0.38$), while college attainment ($r = 0.79$) and poverty rate ($r = -0.74$) are more than twice as predictive.

AI Debate Coach Evaluation | *CRA UR2PhD Research Program* Feb. 2026 – Apr. 2026
Undergraduate Researcher, Advisor: Dr. Yan Ma *Kean University*

- Built an OpenAI supervised fine-tuning pipeline for **GPT-4o mini**, preparing 120 JSONL debate conversations comprising 960 messages across 75 multi-turn and 45 single-turn examples.
- Implemented a 3-condition evaluation framework comparing baseline, prompt-engineered, and fine-tuned debate agents through participant surveys and GPT-4o scoring of sycophancy resistance, rebuttal precision, and cognitive friction.
- Engineered the Node.js/Express study platform with SSE streaming, Neon/PostgreSQL persistence, structured surveys, retry handling, and an authenticated analytics and data-export dashboard.

Professional Experience

Independent Web Developer Jan. 2026 – Present
Freelance Full-Stack Web Development *Sayreville, NJ*

- Independently delivered 5 production websites for 5 paying business clients and built the WiCyS Kean Chapter site using Next.js, TypeScript, PostgreSQL, Supabase, and Cloudflare Workers; 2 client sites include live payment functionality.
- Selected work: daysicecream.com (4,000+ monthly visitors) | trainingaio.com (booking platform, delivered in 2 weeks) | tedishairstudio.com | deafsharkcoffee.com | adzio.io | wicyskean.me.

Computer Science Tutor (Code Samurai) Jan. 2026 – Present
Kean University *Union, NJ*

- Selected by CS faculty to provide one-on-one Database Management Systems tutoring, coaching undergraduates through relational schema design, SQL queries, and step-by-step debugging.

Data Visualization & Analytics Intern July 2026
Group Summer Scholars Research Program, Kean University *Union, NJ*

- Lectured daily, independently developed 8 hands-on labs, and mentored 5 high school students under *Dr. Ching-yu Huang*; all 5 completed database-driven websites and presented individual research posters.